

Cyber Security Assignment

Group-15

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1. INTRODUCTION

The digital revolution has a significant role in transforming all the parts of Cyberia, such as improving life in all aspects, proper and sustainable use of resources, and success of economic prosperity. However, this technology has brought a kind of risk and has become a major concern at all local and international levels. Although it is impossible to provide 100% security, the Government of Cyberia must work cooperatively (publicly & privately) to provide a secure cyberspace and create an appropriate environment without any threats. In addition, there is no standard solution that can be used in all sectors, but a suitable solution must be provided for each system that ensures resilience and covers all risks & threats. So, this paper focuses on developing an NCSS for Cyberia to cover all necessary aspects.

1. 1.The scope of the strategy

This strategy includes all vital sectors and associations such as the public and private sector, academia, civil society and the broader population in Cyberia. The strategy also includes Cyberian Government's policy to offer an integrated and compelling image to deal with stakeholders. This strategy defines the recommended procedures for all sectors down to ordinary citizens. These steps raise awareness of cybersecurity which in turn is of great interest to Cyberia. Here, cybersecurity is the assurance with specific actions to protect both critical infrastructure (CI) and critical information infrastructure (CII) of public and private organisations, and systems in Cyberia, all information exchanged within the systems, and all services exported to beneficiaries. In other words, it protects cyberian cyberspace from cyber threats.

2. STRATEGIC CONTEXT

The national cybersecurity strategy aims to provide and implement a reliable, confidential, and available infrastructure for Cyberia and its Government and citizens even under contingency circumstances. As the strategic context, Cyberia needs to assess the threats and vulnerabilities and the capabilities to develop against their growth. This cyber approach should be aligned with the pillars of the national security strategy and protect national values, even the individuals. Eventually, it will lead to having a cybersecurity culture for Cyberia.

Threats

In this strategy, we face diverse threats, but probably here, we can mention the most crucial type that is cybercrime. Since the biggest problem in the country is the smuggling of weapons and drugs and the criminals can exploit the darknet for their activities, therefore defence against this type of threat will be a high priority strategy for Cyberia. As mentioned, the criminals are using anonymous technologies and devices as tools while performing their criminals' interactions. On the other hand, threats such as fraud, extortion, and ransomware most likely will be increased by organized criminal groups that most of them are located in Eastern Europe[1]. They are responsible for developing and spreading the malware within the countries that have weak cyberspace in security aspects.

Vulnerabilities

Previously, cyberspace was enclosed within the devices such as computers, laptops, mobile devices that were connected to the networks and internet. Nowadays, we have cyberspace in our detailed and daily life, and it is increasing obviously. As a result, the increasing number of devices will be a vulnerability that should be concerned in Cyberia to protect its core systems such as interconnected infrastructure in the society, health, and welfare.

As another type of vulnerability, the weaknesses in compliance could be presented while the victims do not have sufficient awareness and compliance. However, having systematic national leadership, it will mitigate the duplicated resources to be able to address the cyber threats and vulnerabilities while all national stakeholders are cooperating. All the stakeholders, such as individuals, organizations, governments, play a critical role in determining the strategic context and remediate the vulnerabilities to end with it toward establishing a safe and secure cyber environment for Cyberia.

Risk management

The cyber environment always will be compromised by threats and will have identified vulnerabilities. Therefore, the vital point is that we always consider risk management and identify the national needs relevant to the security aspects and establish an information security management system (ISMS). This shall be aligned with the national risk management. In contrast, security risk management is about the continual process and efforts that analyze the possibilities and their consequences and what should be the plan to prevent that risk from happening. Eventually, one of the challenges that Cyberia will face is to have the defense capabilities and agile approaches that should be developed while they can mitigate the increasing number of threats and vulnerabilities within the Cyber era.

3. CYBERSECURITY ENDS

3. 1.Vision, Objectives and Principles

As a country like Cyberia, with its prominent geographical location, important resources and critical infrastructure, Cyberia's vision in 2026 is resilient and secure in cyberspace to protect its critical infrastructure(CI) and critical information infrastructure(CII) against cybersecurity risks in all situations, and enhancing confidence and wealth in the digital world.

To accomplish this vision, the Cyberia strategy includes the following objectives:

- Protect the Cyberia against rising cyber threats and crime
- Secure and resilient cyberspace for tackling cyber attack and be able to ensure secure business.
- Respond effectively to incidents and ensure Cyberia data, networks, and systems are protected and resilient.
- Assist in building stable, open and vibrant cyberspace to ensure freedom, equality and support open societies so that the populace can work safely in cyberspace.
- Improve skills and ability of citizens, businesses and both public and private sectors to defend themselves.
- Having cutting-edge analysis and capabilities to facilitate Cyberia to overcome all cyber security threats and challenges.

To support these objectives, The Cyberia strategy includes the following principles:

- All efforts and policies will be directed to the needs of the citizens' protection and national prosperity.
- Cyber attack will be handled as seriously as would a similar conventional attack and defended as necessary as possible.
- Core values such as democracy, liberty, human rights, freedom of expression, the rule of law, open and accountable institutions and governments will be vigorously preserved and developed.
- Cyberian citizens' privacy will be preserved and protected.
- Working in partnerships includes individual citizens, public sector, institutions and organisations.

3. 2. Stakeholders, roles and responsibility

Individuals

Citizens, employees and consumers should take all reasonable steps to protect their hardware, devices, data, software and systems that make them affordable in their professional and personal lives for flexibility, independence and convenience[1].

Businesses and organisations

Businesses, both public and private sectors organizations, and other institutions have an essential responsibility to safeguard their digital assets. Consumers, citizens and society should converge on business and organization policies that cover all reasonable actions to preserve their data and develop resilience. Businesses and organisations are liable for the consequences of cyber-attack if it happens in their environment[1].

Government

The essential responsibility of the government is to protect the country from invasion by other states, protect its citizens and economy from harm, and establish domestic and international structures to protect its interests, protect fundamental rights, and bring criminals to justice. The government also has a significant responsibility to implement defences for its information assets, advise and inform citizens and organisations to protect themselves online, and ensure national resilience throughout the administration process, necessary services and functions[1].

The Government will play a vital role in national cyber security over the next five years. The Government should utilise all available means, such as GDPR (General Data Protection Regulation), to push up cybersecurity norms across the economy. Coordination with intelligence agencies, the police and the National Crime Agency, as well as international partner agencies, will expand their efforts to disrupt identified cyber activities by cyber criminals. Moreover, the Cyberia Government should build a single, central body for national cyber security. This central body will examine cyber threats and provide cyber security expertise to assist the Government's attempts to promote innovation, support growing cyber security management, and stimulate improvement of cyber security skills[1].

4. PRIORITIES OBJECTIVES

The NCSS is the foundation for a coordinated national and globally harmonious approach to protecting our critical infrastructure (CI) and critical information infrastructure (CII) against cyber threats. Based on Global Cybersecurity Agenda (GCA) of ITU, “ITU NATIONAL CYBERSECURITY STRATEGY GUIDE” and “GUIDE TO DEVELOPING A NATIONAL CYBERSECURITY STRATEGY”[2]–[4], this strategy only covered three strategic pillars or work areas instead of five due to limitation of words and pages.

Three strategic pillars are: **Legal Measures; Technical and Procedural Measures; and Organizational Structures**

4. 1. Legal Measures

It includes the establishment of laws to prevent and prosecute cybercrime. Unavoidably, the steps will depend on national circumstances and local requirements. Establishing and modernizing laws, procedures and policies to deter, prevent, respond and prosecute cybercrime are the fundamental components of the Cyberian NCSS. As cybersecurity becomes a global challenge, the lack of national and regional cybercrime laws will break Cyberia's internal and global cybercrime detection, prosecution and prevention capabilities. The NCSS includes focus areas[4] or actions[3] below as required for creating cybercrime legal measures:

Action 1: Cybercrime Legislation

It includes the formulation of legislation that precisely describes forbidden cyber-activity with aims to reduce online crime, and harmonised with global conventions to align with international cooperation such as ITU Toolkit to get help and addresses jurisdictional and evidentiary issues. Additionally, collaboratively harmonious laws strengthen cybersecurity as it helps Cyberia to increase its capacity to prevent, deter and prosecute cybercrime. The Cybercrime act should be evaluated by all ministries, Legislative Committees, both private and public local sector, academics, interested citizens and willing foreign governments who may be interested in it, so that all-important approaches are covered.

Action 2: Government Legal Authority

The objective is to ensure that governments have adequate legal authority to protect cyberspace in the public interest. Government will attempt the following actions[3] to achieve the aim :

- Design cybersecurity organization structures including the national cyber security centre (NCSC);
- Outlines the legal basis for creating a national CERTs/CIRTs/CSIRTs, For example, the Act defines the powers to shut down a CI or CII during risk of a cyber-attack;
- Provides the support for increasing cybersecurity skills, education and awareness to a range of stakeholders (e.g., judges, lawyers, prosecutors, law-enforcement officials and forensic specialists);
- Defines the legitimate and operational basis for an integrated and fully regulated public-private partnership (PPP) to enhance the security of CI and CII. PPP is the foundation for efficiently protecting CI, managing security risks in both short and long term, and building trust between industry and government;
- Fosters innovation in cybersecurity to ensure that national research attempts support the objectives of the NCSS;
- Grants the power to appropriate authority to engage in international cooperation, dialogue, and coordination activities focused on cybersecurity, such as enabling cross-border investigations and prosecutions, mutual legal assistance, extradition and handling of digital evidence.

4. 2. Technical and Procedural Measures

This priority aims to examine and address the requirement to design organisational structures at local and national levels to facilitate communication, information exchange and the recognition of digital credentials across jurisdictions. The structures would help to create a generic and common digital identity system within the following actions:

Action 1: National Cybersecurity Framework

The objective of this action is to produce a framework that outlines obligatory security models and provides guidance on risk management, compliance and assurance. The Cyberia National Security Framework defines minimum security criteria such as core security values for stakeholders to comply with national cybersecurity requirements. Stakeholders may use internal auditors or external auditors to comply with a central organization's

minimum security standards. However, the framework does not present complete technical guidance on particular ICT systems.

Based on ISO/IEC 27000 Series of standards[5], the NCSS identifies the following policy goals [3] as important components for National Cybersecurity Framework: “

- Governance and Risk Management
- Information Security and Assurance
- Protective Marking and Asset Management
- Staff Vetting and Clearance
- Physical and Environmental Security ”

Action 2: Secure Government Infrastructure

This action aims to spread awareness about relevant risks, preventive measures, and effective response in government departments and agencies. The Government of Cyberia has a significant responsibility to protect its critical information infrastructure. Although the Cyberia Government owns and operates a few CII, Government needs to assign considerable importance to protect both CI and CII. Therefore, the Government needs to ensure secure cyberspace. For instance, the development of security cyberspace technology should be included in the Government’s procurement process. The NCSS classifies the following actions[3] as essential for securing cyberspace: “

- Create and enforce a cybersecurity risk management process across government ministries and agencies;
- Create a vulnerability management process for all government cyber systems;
- Secure government local area networks;
- Create and enforce a staff vetting and clearance scheme;
- Define and enforce a robust government Authentication Framework;
- Make cybersecurity a national, local and regional government accountability;
- Create and enforce a formal information or data classification for sensitive data;
- Improve security in government outsourcing and procurement through vetting of suppliers, incorporation and enforcement of security clauses in contracts.”

Action 3: Critical Information Infrastructure Protection

The objective of this action is to produce a means for tracking and securing vulnerabilities, and developing attack attribution and prevention capabilities. Vulnerability is a weakness or a hole in an application or system, which could be a design error or an implementation bug that allows an attack or threat to harm an application's stakeholders, such as breaking a system’s confidentiality, integrity and availability[6]. Most cyber-attacks commonly happen for poor technical designs or exploitation of the disgruntled worker and unfixed vulnerabilities. The consequence of exploitation of vulnerabilities depends on the value of the CIIs, which inescapably stores valuable information. The NCSS classifies the following essential actions [3] to fix and decrease vulnerabilities and threats: “

- Create a process for national vulnerability assessments to help understand the potential consequences of threats and vulnerabilities;
- Designate important systems as critical information infrastructure and enforce an accreditation regime around them. For example, no system will connect to critical infrastructure without a penetration test and other assurance activities;
- Enhance law enforcement capabilities in the investigation, prevention and prosecution of cybercrimes;
- Require the use of evaluated software products;
- Prioritize national cybersecurity research and development activities;
- Assess and secure emerging systems; and
- Participate in international efforts to improve the security of Internet protocols and routing technologies.”

4. 3. Organisational Structures

This priority aims to address the requirement to design organisational structures and strategies to detect and respond to cyber threats against critical infrastructure(CI). The NCSS includes focus areas or actions below as essential for creating proper public and private organizational structures, roles and responsibility, and policies on cybercrime:

Action 1: Government's Cybersecurity Role

The objective of this action is to highlight Governments' and relevant stakeholders' responsibility for securing cyberspace. Everyone is responsible for securing cyberspace because countermeasures will be effective only if all relevant stakeholders perform their role. However, the Cyberian government has an overall responsibility to protect the CII and the CII in the public interest by establishing an appropriate national structure. The purpose of a coherent organizational structure publicly, privately and globally is to protect classified data and networks against cyber attacks. The government is also responsible for communicating about national priorities in the private sector to ensure adequate protection of CIs in private hands such as banking, transport and telecommunications[3]. To handle these challenges, Cyberian government may also appoint a senior associate as national cybersecurity coordinator with the responsibility to organise a cross-government agenda to label the priority actions of this strategy. The Coordinator should have effective communication and coordination with the head of government and sufficient staff and financial resources (i.e. dedicated budget) to coordinate inter-government activities at a strategic level[3].

Action 2: National Incident Management Capacity

This action aims to create a national point-of-contact for round the clock cyberspace analysis, significant incident response and information sharing. Knowledge of timely detection, communication and recovery from various deliberate incidents can often mitigate the damage caused by malicious exercise. In general, most of the infrastructure is owned by the private sector, and they have better skills to manage deliberate incidents. So, Inescapably, the government needs to engage with them to organise a national response and ensure that CIs owner's and operator's systems are resilient to attacks by making legal and regulatory incentives. The NCSS classifies the following essential actions [3] considering cyber incident response: “

- Build National Computer Incident Response Team (N-CIRT) at the National Cybersecurity Agency;
- Establish a public-private framework for responding to major cyber incidents;
- Encourage development of business continuity and disaster recovery capacity;
- Develop strategic and tactical cyber attack and vulnerability assessment capacity;
- Encourage the development of private sector capacity to share status information about the health of cyberspace;
- Coordinate processes for voluntary participation in the development of national public-private continuity and contingency plans;
- Create a Cyber Warning and Information Network at Cybersecurity Agency;
- and Develop mechanism to share information about cyber attacks, threats and vulnerabilities with the public and non-governmental bodies locally and globally.”

Action 3: Cyber security Skills and Training

The objective of this action is to produce a means programmes to enhance the capability of cybersecurity professionals in technical, managerial, and information maintenance fields. The NCSS identifies the following necessary activities [3] considering the enhancement of the capability of cybersecurity professionals: “

- Adopt a national Cybersecurity Skills Framework;
- Create a continuum of cybersecurity job descriptions;
- Identify commercially available cybersecurity certifications;
- Deliver or manage commercial delivery of training or certification examinations;
- Periodically measure Cybersecurity skills and training levels;
- Invest in mainstream cybersecurity education and research;
- Build cybersecurity capacity of national companies; and
- Work with global partners such as IMPACT to coordinate cybersecurity training.”

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